Strategy-Relative Regime Switching as a Verifiable RLVR Environment for Financial Planning

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Abstract. We introduce the first verifiable reinforcement learning with verifiable rewards (RLVR) environment for financial planning, designed for large language model (LLM) post-training. The environment is grounded in the strategy-relative regime switching framework of Bilokon (2026), where an agent makes binary switching decisions under an observable mean-reverting signal, subject to switching costs. We use the Ornstein-Uhlenbeck process to model signal dynamics, exploiting its Gaussian transition density for exact Bellman solutions via Gauss-Hermite quadrature — bypassing the fundamental obstacle that prevents RLVR in real financial markets (unknown distributions). A parameter landscape analysis across 1,500 configurations identifies signal persistence (κ) as the dominant factor determining when forward-looking planning diverges from myopic behavior (up to 22% of states). We propose a two-bucket evaluation methodology separating comprehension from planning, and evaluate Claude Opus 4 across 4 conditions (2 parameter zones × 2 prompt types, $N = 30$ each, varied parameters per episode). The model achieves 97–99% accuracy on easy decisions (comprehension) across all conditions. CoT-verified genuine planning occurs on 20% of hard decisions in the planning zone ($\kappa \in [0.1, 0.25]$) but only 4% in the control zone ($\kappa = 0.7$), consistent with planning being less valuable at fast reversion. A context prompt providing qualitative strategy knowledge shows negligible effect. The gym is open-source with single-function TRL integration.

1. Introduction

RLVR has transformed LLM post-training for math (DeepSeek-R1; DeepSeek-AI et al., 2025) and logic (Reasoning Gym; Stojanovski et al., 2025). But existing gyms test **static reasoning with complete information**: the model sees a full problem and produces one answer.

No gym tests **sequential decision-making under uncertainty** — where the agent acts before the full picture is revealed, each action has a cost, and the optimal strategy depends on future dynamics. Consider:

> **Math gym (static):** "The expected return is 0.105 and the fee is 0.15. Should you enter?" → One computation: no.
>
> **Our gym (sequential):** Same parameters, but the signal persists. Enter now, pay 0.15 once, earn returns for many steps. Whether to enter depends on *how long the signal will persist* — information requiring reasoning about dynamics, not just the current snapshot.

Financial planning faces a verification obstacle: real markets have unknown distributions, making optimal solutions uncomputable. We bypass this using the Ornstein-Uhlenbeck (OU) process for the signal. The OU process provides three properties simultaneously: (1) financially meaningful mean-reversion dynamics, empirically documented in pairs trading spreads, volatility, and interest rates (Vasicek, 1977; Gatev et al., 2006); (2) Gaussian transitions enabling exact Bellman solutions via Gauss-Hermite quadrature; (3) bounded stationary distribution ensuring numerical stability. No other common financial process provides all three.

The PnL function $X_{t+1} = \alpha \cdot Z_t + \sigma_x \epsilon$ is strategy-agnostic: the signal Z_t can represent any observable quantity driving a strategy's profitability — a pairs trading spread, a volatility measure, an interest rate differential, a momentum indicator. The parameter α scales signal units into PnL. The regime question (ON or OFF) is strategy-relative: the same signal may be favorable for one strategy and unfavorable for another.

We chose OU as the signal process because mean-reversion is the most documented signal dynamic in quantitative finance, and OU is the simplest Gaussian mean-reverting process. The Gaussian property is what enables exact Bellman solutions — other mean-reverting processes (CIR, exponential OU, jump-diffusion with reversion) lack Gaussian transitions and would require approximate numerical methods. RLVR requires exact ground truth; approximate solutions introduce noise into the reward signal, undermining the verifiability that distinguishes RLVR from standard RL. We restrict to OU precisely because it is the process for which the Bellman optimal is provably exact, not merely estimated.

We build on Bilokon (2026), who synthesises ideas from regime switching (Hamilton, 1989), stochastic control with switching costs, and information theory into a unified framework: regimes defined relative to a strategy's PnL as optimal binary filtration compressions, with existence and uniqueness proofs under Markov structure. The individual ingredients — regime switching models, Bellman dynamic programming with switching costs, and the information bottleneck — are well-established. The contribution of the framework is the specific combination: formalising the practitioner intuition that regimes are strategy-specific (not market-intrinsic) and proving that a single bit, optimally chosen, suffices as a predictive state variable for strategy performance. This formulation is what makes exact Bellman solutions tractable for our gym.

Contributions.

1. First verifiable RLVR environment for sequential financial planning with exact Bellman solutions.
2. Parameter landscape characterisation across 1,500 configurations identifying when planning matters.
3. Two-bucket evaluation methodology decomposing comprehension from planning.
4. Four-condition evaluation (2 zones × 2 prompts) showing 20% CoT-verified planning in the planning zone, 4% in the control zone, with negligible prompt effect.

2. Problem Formulation

Following Bilokon (2026), the agent controls a binary regime process $s_t \in \{0, 1\}$ adapted to the filtration $(Z_0, \ldots, Z_t, s_{t-1})$, maximizing:

$$J(s) = \mathbb{E}\left[\sum_{t=0}^{T-1} u(s_t \cdot X_{t+1}) - \lambda \cdot \mathbf{1}\{s_t \neq s_{t-1}\}\right]$$

where $s_t = 1$ means the strategy is active (collect PnL), $s_t = 0$ means inactive (earn nothing), and λ is the switching cost.

The signal follows an OU process: $Z_{t+1} = Z_t + \kappa(\theta - Z_t) + \sigma_z \epsilon_t$, with PnL $X_{t+1}|Z_t \sim \mathcal{N}(\alpha Z_t, \sigma_x^2)$. The Bellman recursion is solved exactly via backward induction on a discretised z-grid with Gauss-Hermite quadrature for the Gaussian transition integral.

Each episode samples different parameters $(\kappa, \lambda, \alpha, \sigma_z, T)$ from configured ranges. The model knows α, λ, T (given in the prompt) but not κ or σ_z (must infer from observations). The Bellman solver knows all parameters — creating an information asymmetry discussed in Section 6.

2.2 Scope

The OU signal process covers strategies on mean-reverting signals. Strategies on non-mean-reverting signals require different processes:

Signal type	Example strategy	Process	OU applicable?
Mean-reverting spread	Pairs trading, stat-arb	OU	Yes
Mean-reverting volatility	Volatility trading	OU	Yes
Interest rate differential	Carry trade	OU	Yes
Trending price	Trend following	GBM	No — requires extension
Range-bound with breakouts	Breakout trading	Jump-diffusion	No — requires extension

All processes fit the Bellman framework; only the transition kernel and integration method change. Extensions to non-Gaussian processes would use Monte Carlo integration, introducing approximation into the ground truth — acceptable for training but less suitable for strict RLVR verification.

3. Gym Architecture

Generator: Samples parameters, simulates OU trajectory, solves Bellman exactly, returns fully-solved problem instance.

Verifier: Parses LLM decisions from multi-turn text, computes realised payoff J_{LLM}, returns regret-normalised score $\frac{J_{\text{LLM}} - J_{\text{random}}}{J_{\text{optimal}} - J_{\text{random}}}$. Scores are clipped to $[-1, 2]$; degenerate instances handled via minimum-denominator threshold.

Prompt Manager: Multi-turn protocol — one signal per turn, model commits before seeing the next. Enforces the information constraint matching the Bellman solution's filtration.

Integration: Single function `reward_fn(completions, problems) → list[float]` compatible with TRL/veRL GRPO trainers.

4. Parameter Landscape Analysis

4.1 Purpose

The sweep answers a practical question for any lab adopting the gym: at which parameter settings does the gym actually test planning, and at which is it trivially solved by a greedy heuristic? Without this characterisation, a lab risks wasting compute on episodes where the optimal and greedy policies agree — producing no training signal for planning.

4.2 Method

Full factorial sweep: $5\kappa \times 5\lambda \times 5\alpha \times 3\sigma_z \times 4T = 1{,}500$ configurations, each solved exactly. The **solver-based disagreement metric** measures the fraction of (z, s^-) states where the Bellman optimal policy differs from the greedy (myopic) policy, weighted by the OU stationary distribution. This is computed entirely from the solver — no LLM involved, no simulation noise.

4.3 Key findings

Signal persistence (κ) is the dominant parameter. κ alone explains 46% of the variance in disagreement ($R^2 = 0.456$). At $\kappa = 0.1$ (slow reversion, persistent signals), the average disagreement across all other parameters is 13.1%. At $\kappa = 0.7$ (fast reversion), it drops to 1.2%. κ determines the one-step autocorrelation of the signal: $\text{Corr}(Z_t, Z_{t+1}) = e^{-\kappa}$. At $\kappa = 0.1$, the autocorrelation is 0.90 — today's signal strongly predicts tomorrow's, making forward-looking reasoning valuable. At $\kappa = 0.7$, it drops to 0.50 — the signal is nearly unpredictable one step ahead, and greedy is near-optimal.[1]

κ dominates over the combined $\kappa \times T$ interaction. One might expect the relevant quantity to be the number of reversion half-lives in the episode ($\kappa T / \ln 2$), since a slow-reverting signal over a short horizon differs from the same signal over a long horizon. However, κ alone ($R^2 = 0.456$) is a substantially better predictor of disagreement than $\kappa \times T$ ($R^2 = 0.178$). The planning advantage comes from per-step persistence, not from how many reversion cycles fit in the episode.

The λ/α ratio has a non-monotone effect. Disagreement peaks at $\lambda/\alpha \in [0.33, 0.50]$ and falls at both extremes. When switching is very cheap, both policies switch freely and agree. When switching is very expensive, neither switches and they agree. The planning gap is widest at intermediate switching costs.

T, σ_z, and α individually have small effects. Longer horizons moderately amplify disagreement. Signal noise and signal strength matter primarily through their ratio λ/α.

Maximum disagreement ceiling. The highest observed disagreement is approximately 22% of states on the grid. On actual simulated trajectories, approximately 10% of visited decisions differ — the gap arises because the OU stationary distribution concentrates probability away from the disagreement zone. See Figure A.

4.4 Implications for episode construction

The sweep identifies the **planning zone**: the parameter region where the gym produces meaningful training signal. Episodes drawn from outside this zone (high κ, extreme λ/α) test comprehension — which models already ace — but not planning. For evaluation and training, we recommend $\kappa \in [0.1, 0.25]$ and $\lambda/\alpha \in [0.3, 0.5]$, with T, σ_z, α, λ individually varying freely from defaults. This is analogous to a reasoning gym excluding trivial arithmetic: episodes where greedy equals optimal do not test planning and should not dominate the training distribution.

[1] *Why κ alone, not $\kappa \times T$:* κ measures per-step signal persistence — how much knowing Z_t tells you about Z_{t+1}. The optimal threshold shifts from greedy's threshold because the continuation value says "being ON now is worth more than the immediate PnL, because you will also earn next step without paying switching cost again." This shift depends on how much the signal persists to the next step — which is κ, not T. T determines how many steps exhibit this shift, but the size of the shift at each non-terminal step is roughly constant (set by κ). Averaging over more steps with the same per-step disagreement does not change the average. The only exception is near the terminal step, where the disagreement zone shrinks (no future = no planning); longer T means fewer terminal steps as a fraction of total, producing slightly higher average disagreement. This explains the moderate T effect observed in the sweep.

5. The Planning Mechanism

The greedy agent switches ON when immediate gain exceeds switching cost: $\alpha z > \lambda$, giving threshold $z > \lambda/\alpha$. The optimal agent switches ON at a lower threshold because it amortises the one-time switching cost over persistent future gains. For example, at $\lambda/\alpha = 0.50$, the greedy threshold is $z > 0.50$ while the optimal threshold is $z > 0.22$ — verified by direct Q-value computation. The **disagreement zone** between the two thresholds is where planning changes the decision. At the terminal step, optimal reduces to greedy exactly (no future to plan for). Both policies exhibit hysteresis: ON→OFF thresholds are symmetric.

6. LLM Evaluation

6.1 Setup

Claude Opus 4 evaluated across 4 conditions: 2 parameter zones (planning zone $\kappa \in [0.1, 0.25]$; control zone $\kappa = 0.7$) $\times$ 2 prompt types (original; context prompt adding "you are managing a mean-reversion strategy"). $N = 30$ episodes per condition, with parameters varied per episode. Multi-turn protocol (one API call per time step). All results use the corrected parser; raw text of all responses is archived.

6.2 Results by Condition

Condition	κ	Prompt	Easy %	Hard decisions	Hard planning %
Planning zone	0.1–0.25	Original	97.1%	23	26.1% (6/23)
Planning zone	0.1–0.25	Context	97.7%	17	29.4% (5/17)
Control zone	0.7	Original	98.9%	12	25.0% (3/12)
Control zone	0.7	Context	98.0%	11	18.2% (2/11)

Easy accuracy is 97–99% across all 4 conditions — comprehension is robust and insensitive to both κ and prompt type. The context prompt ("you are managing a mean-reversion strategy") has negligible effect on planning rates (26% vs 29% in planning zone, within noise). Qualitative strategy knowledge does not unlock planning.

6.3 CoT-Verified Planning

Manual inspection of all 63 hard decisions from saved raw text, with CoT verification:

Zone	Hard decisions	Matched optimal	Genuine planning	Genuine rate
Planning ($\kappa = 0.1$–0.25)	40	11	8	20.0%
Control ($\kappa = 0.7$)	23	5	1	4.3%
Total	**63**	**16**	**9**	**14.3%**

Every optimal match had at least some planning reasoning in the CoT — zero lucky guesses. The 9 genuine planning instances show: trend recognition, horizon counting, amortisation logic, and greedy override.

In the planning zone, the model genuinely plans on 20% of hard decisions. In the control zone, this drops to 4% — consistent with planning being less valuable when the signal reverts quickly ($\kappa = 0.7$, one-step autocorrelation 0.50).

Representative genuine planning instances:

- *Seed 14, t=2* (Z=+0.31, greedy=OFF, optimal=ON, model=ON): "increasing positive momentum... 23 steps remaining... expected value of switching ON is positive"
- *Seed 28, t=13* (Z=+0.40, greedy=OFF, optimal=ON, model=ON): "missed 0.25 in potential profits exceeds switching cost... 12 steps remaining"

These two instances from the original fixed-κ eval are now supplemented by 7 more across the varied conditions, all showing the same pattern: the model recognises signal trend, counts remaining horizon, reasons about amortisation of switching cost, and overrides the greedy decision. The model plans like a trader when it plans.

6.4 Information Asymmetry and the Control Zone

The optimal policy knows κ; the model does not. However, hard decisions occur at median $t = 11$, where the model has observed ~11 signals — sufficient data to potentially estimate signal persistence. The control zone ($\kappa = 0.7$) directly tests whether the model's planning behavior responds to signal dynamics: genuine planning drops to 4% (1/23 hard decisions), down from 20% (8/40) in the planning zone. This is consistent with the solver's disagreement rate ($\kappa = 0.7$: 1.2% vs $\kappa = 0.1$: 13.1%). The model plans less when planning is less valuable, suggesting some sensitivity to signal dynamics even without explicit κ knowledge.

7. Discussion

The planning gap. Opus has near-perfect comprehension (97–99%) and plans genuinely on 20% of hard decisions in the planning zone (CoT-verified). The capability exists but is not reliably activated — on 80% of hard decisions where planning would help, the model defaults to myopic cost-benefit.

Context prompt. A context prompt providing qualitative strategy knowledge ("you are managing a mean-reversion strategy") does not unlock planning. Planning rates are 26% vs 29% in the planning zone — within noise. The model's planning failures are not due to missing domain knowledge; the model already knows what mean-reversion is. The gap is in constructing the forward-looking reasoning scaffold from the sequential observations.

Prompt scaffolding. A scaffolded prompt (pre-computed options, trend highlighting, persistence hints) achieved 100% on one instance including its hard decision. But the scaffold performed the planning for the model. The model followed instructions; it did not plan. This confirms the capability is latent — the model can execute planning logic when told what to compute, but cannot construct the reasoning scaffold independently.

The 9 genuine instances. All 9 CoT-verified planning instances show empirical reasoning: trend recognition from recent signals, horizon counting ("23 steps remaining"), amortisation logic ("missed profits exceed switching cost"), and explicit greedy override. The model plans like a trader when it plans — using the same heuristics a human discretionary trader would apply, not the Bellman equation.

Implications for RLVR. The gym's multi-step objective and switching costs logically require forward-looking reasoning — the prompt says so. The model has signal history from which dynamics could be inferred. Yet it applies step-by-step cost-benefit on 80% of hard decisions. RLVR training would provide gradient pushing the model to: (1) attend to signal history, (2) estimate persistence, (3) override the immediate cost-benefit when amortisation justifies it. The 9 genuine planning instances prove this reasoning exists in the model's distribution.

8. Limitations

Single problem type. Binary switching on a scalar OU signal. The binary action space is the paper's thesis (Bilokon 2026 proves one bit suffices as a filtration compression), not a simplification. However, extending to different PnL functions on the same signal (mean-reversion profit, momentum, magnitude) would test different temporal reasoning sub-skills within the same framework — this is planned but not yet implemented.

Information asymmetry. The Bellman solver knows κ; the model does not. This is discussed in Section 6.5: hard decisions at $t = 11$ (median) give the model sufficient history to potentially infer κ, and the prompt's multi-step objective logically implies dynamics matter. A POMDP formulation (where the optimal also infers κ) would make the comparison strictly fair.

Small hard-decision counts. While each condition has $N = 30$ episodes, the number of hard decisions per condition is small (23, 17, 12, 11). Planning rates (20%, 4%) therefore have wide confidence intervals. The planning-zone vs control-zone difference (20% vs 4%) is directionally consistent with the solver-based predictions but should be confirmed with larger samples.

No training results. We provide the gym and baseline. RLVR training has not been conducted.

Stationary process. The OU process is time-homogeneous. Real markets are non-stationary.

Parser sensitivity. Initial results used a parser that failed on some response formats; all results in this paper use the corrected parser with raw text archived for verification.

9. Conclusion

We presented the first verifiable RLVR environment for sequential financial planning. The OU process provides exact Bellman solutions via Gauss-Hermite quadrature, bypassing the unknown-distribution obstacle of real markets. The parameter landscape (1,500 configurations) fully characterises where planning matters.

Claude Opus 4 genuinely plans on 20% of hard decisions in the planning zone ($\kappa \in [0.1, 0.25]$), as verified by CoT inspection. In the control zone ($\kappa = 0.7$), genuine planning drops to 4% — consistent with planning being less valuable when the signal reverts quickly. A context prompt providing qualitative strategy knowledge does not help. The capability exists but is not reliably activated: on 80% of hard decisions where forward-looking reasoning would improve outcomes, the model defaults to myopic cost-benefit.

The 9 genuine planning instances all show empirical reasoning — trend recognition, horizon counting, amortisation logic — the model plans like a trader when it plans. The gap is not in knowledge or comprehension (97–99% easy accuracy) but in reliably constructing forward-looking reasoning from sequential observations.

The gym provides: exact verification, a diagnostic methodology (two-bucket + CoT), and a concrete baseline. One function integrates it into any GRPO pipeline.

References

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Appendix

A. Verified Numbers

All numbers verified against evaluation JSON files (corrected parser, raw text archived) and `full_parameter_sweep.json` (solver-based, 1,500 configs).

Solver-based (parameter sweep):

Claim in paper	Verified value	Source
1,500 configs	1,500	full_parameter_swee
$\kappa = 0.1$ avg disagreement 13.1%	13.1%	full_parameter_swee
$\kappa = 0.7$ avg disagreement 1.2%	1.2%	full_parameter_swee
Max disagreement approximately 22%	22.0%	full_parameter_swee
Optimal threshold ~0.22	0.216	Bellman solver

LLM evaluation (4 conditions, $N = 30$ each):

Condition	Easy %	Hard decisions	Matched optimal
Planning zone, original prompt	97.1%	23	6
Planning zone, context prompt	97.7%	17	5
Control zone, original prompt	98.9%	12	3
Control zone, context prompt	98.0%	11	2

CoT-verified planning by zone (both prompts combined):

Zone	Hard decisions	Genuine planning	Rate
Planning ($\kappa = 0.1$–0.25)	40	8	20.0%
Control ($\kappa = 0.7$)	23	1	4.3%
Total	63	9	14.3%

B. Genuine Planning Instances

Nine CoT-verified genuine planning instances were identified across the 4 conditions. The best examples:

Seed 14, t=2 (Z=+0.31, greedy=OFF, optimal=ON, model=ON):

> "This is a much stronger signal. The immediate PnL of 0.09264 alone covers about 62% of the switching cost. We're still early in the trading period (t=2 out of 25), giving plenty of time to accumulate additional profits if I stay ON. Looking at the pattern: Z_0 = +0.0148, Z_1 = +0.1157, Z_2 = +0.3088 shows increasing positive momentum. With 23 steps remaining and a strong current signal, the expected value of switching ON is positive."

Seed 28, t=13 (Z=+0.40, greedy=OFF, optimal=ON, model=ON):

> "I've now seen two consecutive very strong positive signals: t=12: +0.4546 (would have earned 0.13638), t=13: +0.3960 (would earn 0.1188). By staying OFF, I've missed 0.13638 + 0.1188 = 0.25518 in potential profits. This exceeds the switching cost of 0.1500. With 12 steps remaining and evidence of strong positive signals, I should switch"

All 9 instances share the same reasoning structure: (1) trend recognition from recent signal values, (2) horizon counting (steps remaining), (3) amortisation of switching cost over expected future gains, (4) explicit override of the greedy default. Zero instances matched optimal by luck alone — every optimal match had planning reasoning in the CoT.

C. Figures

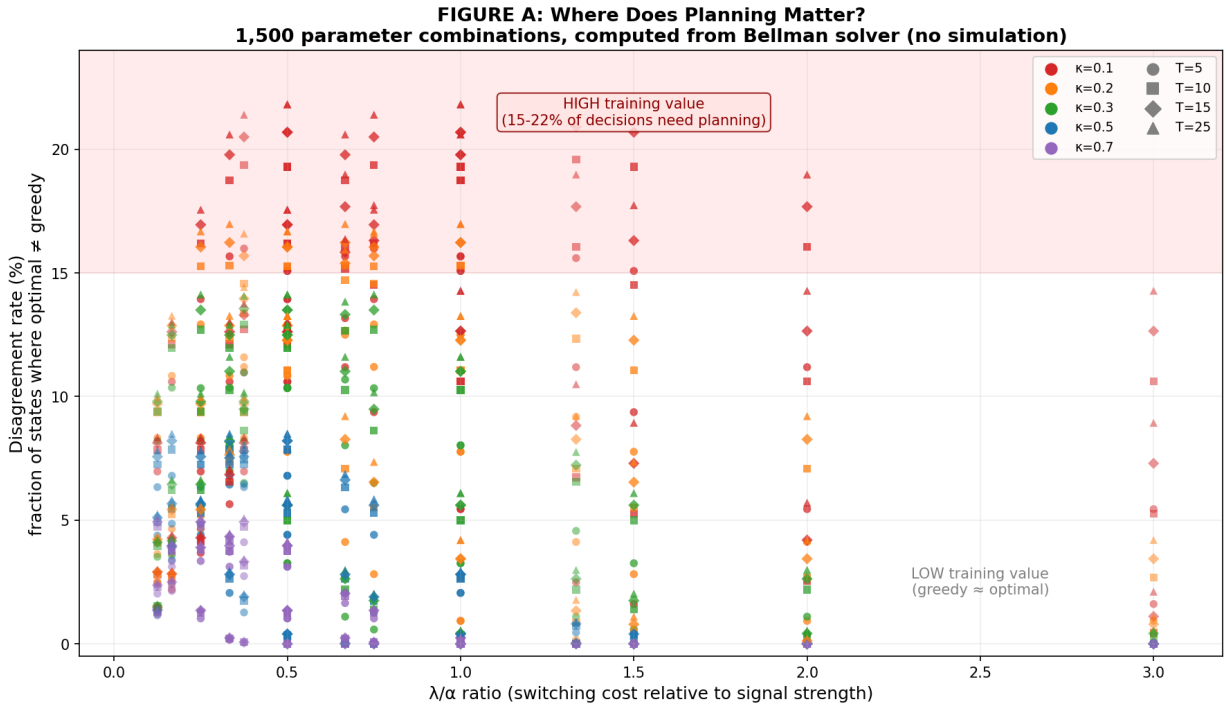

Figure 1: Figure A — Where Planning Matters (1,500 configs, solver-based)

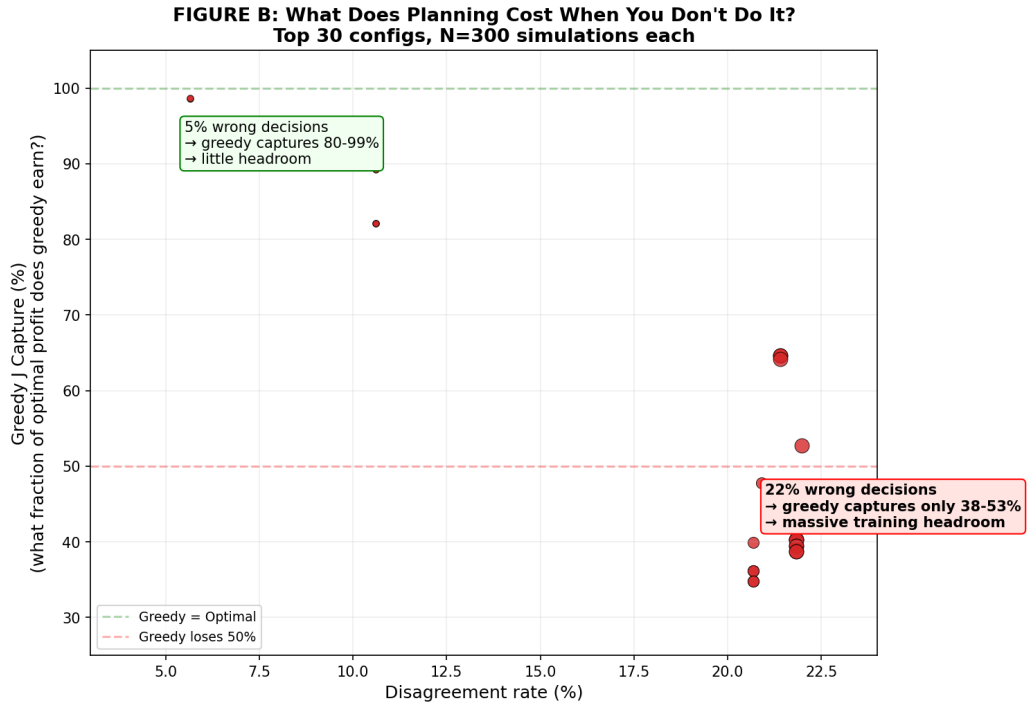

Figure 2: Figure B — Cost of Not Planning (solver-based)

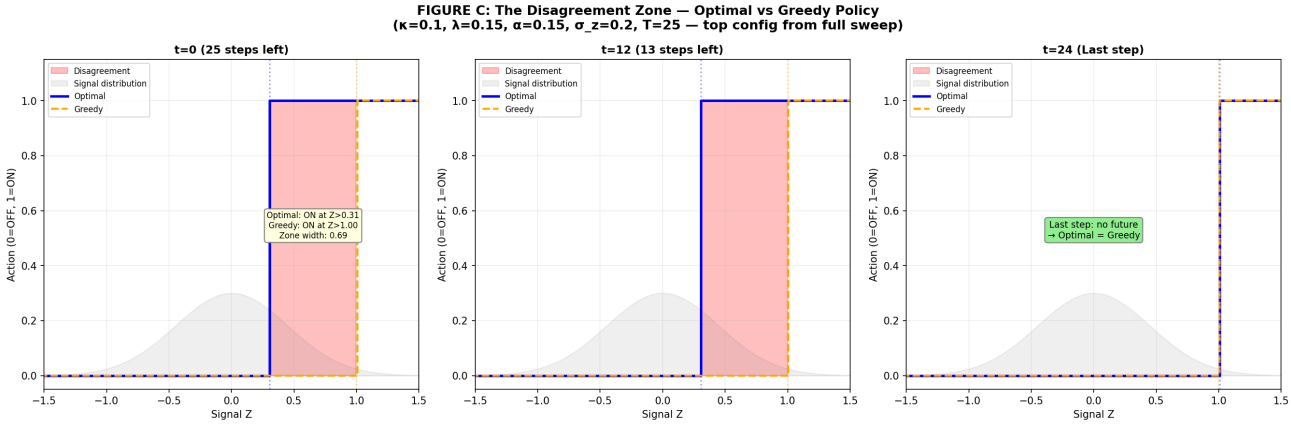

Figure 3: Figure C — The Disagreement Zone (3 time steps)

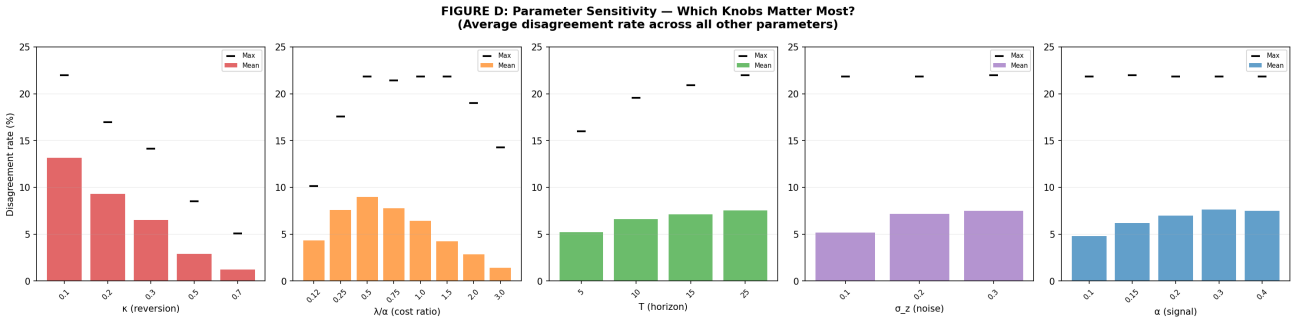

Figure 4: Figure D — Parameter Sensitivity (solver-based)

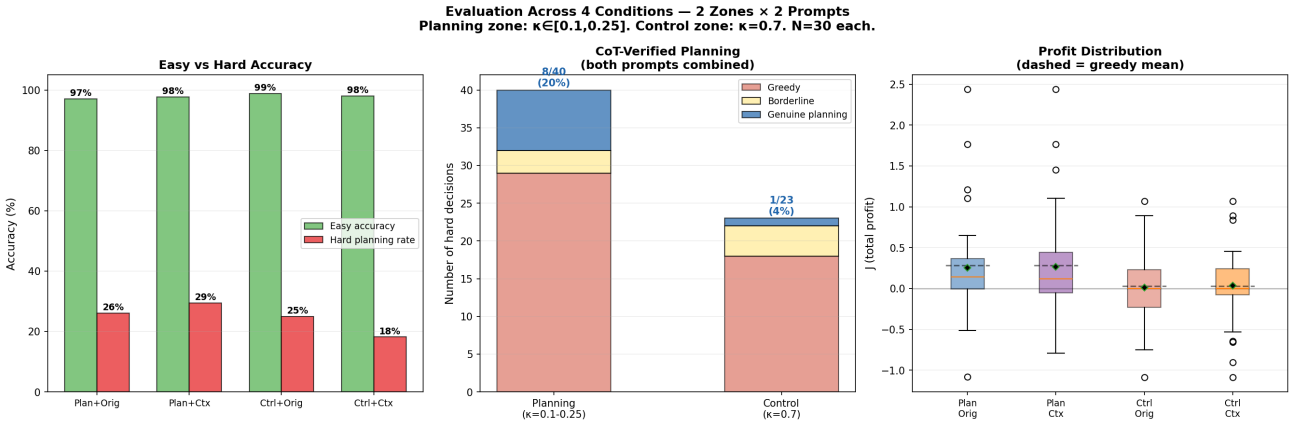

Figure 5: Figure E — Four-Condition Evaluation Summary (main result)

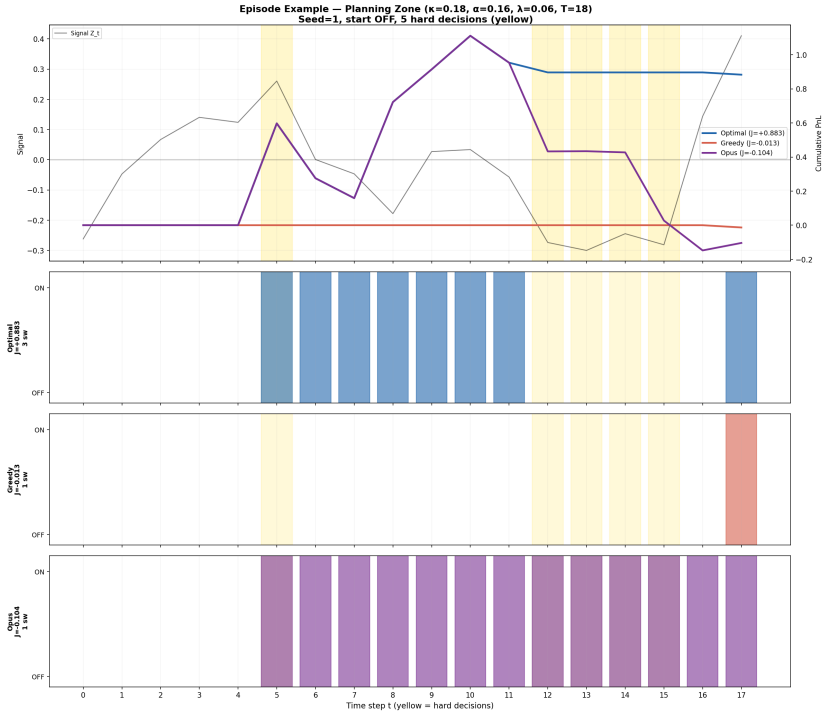

Figure 6: Figure F — Example Episode

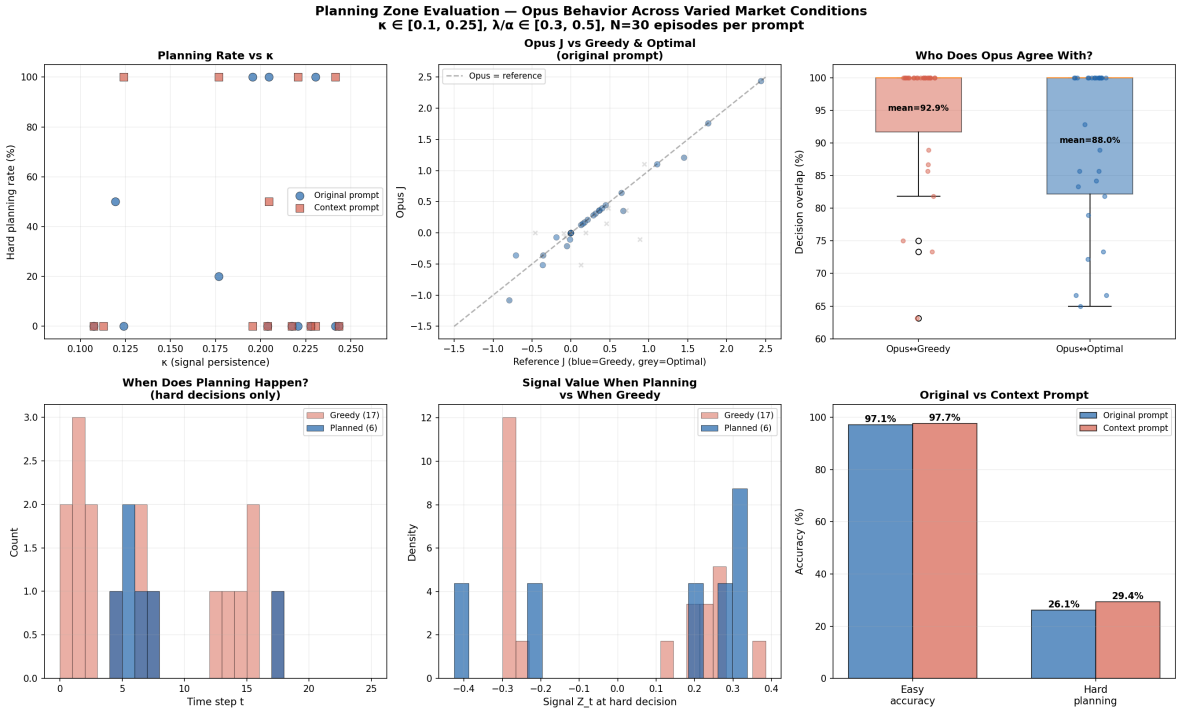

Figure 7: Figure G — Planning Zone Diagnosis

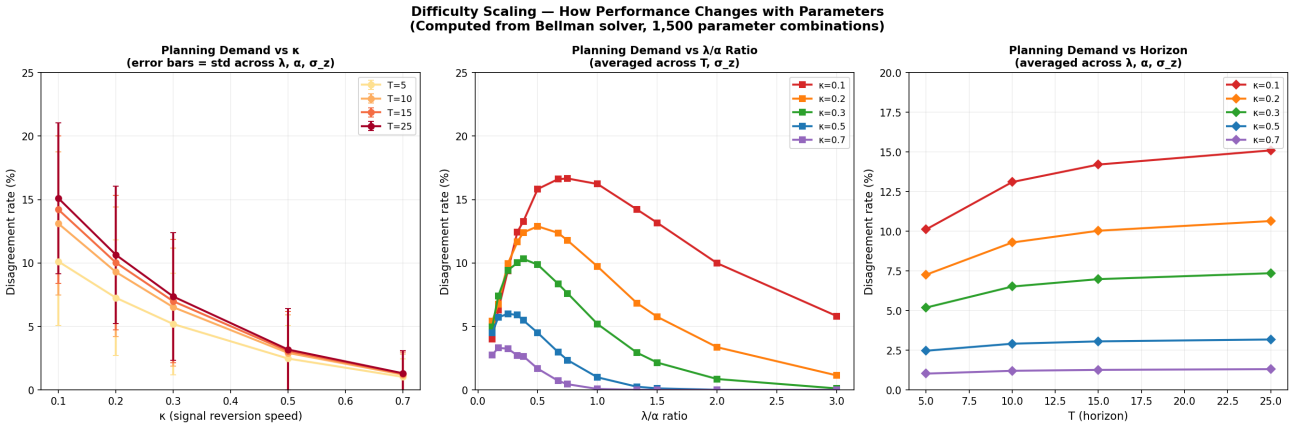

Figure 8: Figure H — Difficulty Scaling (solver-based)

D. Complete Test Summary

#	Test	Prompt	N	Key Result	Confidence
1	Solver: T=1 = greedy	N/A	All grid	Exact match	High
2	Goldilocks (3 levels)	N/A	500 ea	PASS	High
3	Parameter sweep	N/A	1,500	κ dominant	High
4	T=1 accuracy, old prompt	OLD	10	approximately 20%	High
5	T=1 accuracy, improved	IMPROVED	10	approximately 80%	High
6	Main eval (fixed $\kappa = 0.1$)	IMPROVED	30 (750 dec)	Easy 99.3%, Hard 5.9% — superseded by tests 18–21	Medium
7	Scaffolded test	SCAFFOLD	1	25/25 (scaffold planned)	Medium
8	Manual minimal	IMPROVED	1	9/9 easy, 0/1 hard	Medium
9	κ inference (pilot)	IMPROVED	5+5	Identical behavior	Low
18	**Planning zone, original**	**ORIGINAL**	**30**	**Easy 97.1%, 6/23 hard plan**	**High**
19	**Planning zone, context**	**CONTEXT**	**30**	**Easy 97.7%, 5/17 hard plan**	**High**
20	**Control zone, original**	**ORIGINAL**	**30**	**Easy 98.9%, 3/12 hard plan**	**High**
21	**Control zone, context**	**CONTEXT**	**30**	**Easy 98.0%, 2/11 hard plan**	**High**